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From Equality to Excellence: Exploring the Relationship between Gender Equality HR Policies and R&D Intensity

Sung-Tae Lee [1] and Sun-Moon Jung [2,*]

[1] School of Electronic and Electrical Engineering, Hongik University, Seoul 04066, Republic of Korea; innerpeace7654@gmail.com
[2] Dongguk Business School, Dongguk University, Seoul 04620, Republic of Korea
* Correspondence: sunmoonj92@gmail.com

Abstract: In an era where innovation is the key to staying competitive, understanding the role of gender diversity in fostering corporate creativity has never been more crucial. This study examined the relationship between gender equality human resource (HR) policies and a firm's innovation focus, measured by research and development (R&D) intensity and intangible asset intensity. Utilizing a comprehensive survey dataset focused on female workers in Korean listed firms for the period 2020–2022, we found no clear evidence of a positive association between gender equality HR policies such as equal employment opportunities and gender pay equity and both R&D intensity and intangible asset intensity. Furthermore, our analysis revealed that the positive influence of these policies on R&D (intangible) intensity becomes more apparent when the proportion of female employees among the total R&D HR increases. These findings emphasize the critical role that gender-inclusive HR policies play in fostering a culture of innovation within organizations and highlight the importance of considering workforce composition while shaping innovation outcomes. This research contributes to the growing body of literature on the intersection of gender equality, HR practices, and corporate innovation strategies.

Keywords: R&D intensity; gender equality; human resource management; equal employment opportunity; pay gap

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1. Introduction

In an era characterized by rapid technological advancements and dynamic global markets, innovation stands as a cornerstone of corporate success and economic growth. A critical determinant of an organization's innovative prowess is its ability to foster a culture that embraces diversity and inclusivity [1]. Among the various facets of inclusivity, gender equality has emerged as a pivotal factor in shaping the innovation landscape of firms [2]. Gender equality human resource (HR) policies such as equal employment opportunities and gender pay equity have garnered significant attention for their potential to not only rectify historical disparities, but also invigorate the creative potential of a diverse workforce. Understanding the nexus between gender equality policies and innovation, particularly in the context of research and development (R&D) intensity, is a timely and imperative research inquiry. Furthermore, while gender diversity can include a larger spectrum, the present study concentrated on those whose gender identity matched the sex they were given at birth.

Scholarly endeavors have explored the determinants of R&D intensity extensively, identifying factors ranging from financial capability to industry dynamics. However, the influence of gender equality HR policies on this critical facet of corporate strategy remains a relatively underexplored terrain. Existing studies have primarily focused on financial and industry-specific determinants, inadvertently sidelining the broader socio-cultural aspects that underpin organizational innovation [3–5]. Several financial studies have explored gender diversity in boards; however, their focus has been limited to the board of

directors, remaining silent on broader HR policies that affect rank-and-file personnel [1,2]. Furthermore, other studies have implied that female employees may avoid undertaking risky projects due to their risk-averse characteristics [6,7]. Thus, a comprehensive examination of how gender equality HR policies intersect with R&D intensity is needed. This study endeavors to bridge this gap in the literature by examining the relationship between gender equality HR policies and a firm's R&D intensity, and subsequently, the intangible asset intensity.

The motivation for this research is based on the need to understand the nuanced interplay between gender equality initiatives and innovation pursuits in the corporate context. By discerning the impact of gender equality HR policies on R&D intensity, our aim was to provide empirical evidence that not only substantiates the importance of inclusivity, but also underscores its potential as a catalyst for innovation. Furthermore, our research extended beyond the general scope to investigate the impact of workforce composition inside R&D departments, shedding light on the dynamics of gender inclusion in innovation-driven organizations. In particular, we found that the proportion of female workers who benefitted from HR policies had a considerable impact on the relationship between gender equality policies and a firm's R&D intensity. This cross-sectional variation, driven by employee composition, may explain why prior studies have provided conflicting views on the relationship between gender diversity and firm innovation [1,2,6,7].

Our empirical analysis was based on the survey data of HR managers and female workers within Korean listed firms during the period 2020–2022. This survey comprised unique information about the employment practices and career trajectories of female workers, coupled with comprehensive financial information pertaining to R&D activities, providing a powerful context for investigating the interplay between gender equality policies and a firm's innovation focus. By using the survey, we first examined whether gender equality HR policies such as equal employment opportunities for both genders and gender pay parity had a significant positive relationship with a firm's level of R&D investment. We then examined whether the impact of gender equality policies on R&D intensity was more pronounced in firms with a higher representation of females in the overall R&D workforce. Finally, we examined whether gender-inclusive recruiting and equal remuneration practices enhanced the female R&D personnel's dedication, job satisfaction, and performance, thereby strengthening a company's innovation focus.

Our study contributes to the research on gender diversity and corporate investment by exploring HR strategies for rank-and-file employees, beyond the presence of women in top management. While board gender diversity has been linked to higher R&D investment in several studies [1,2], our findings indicate that broad HR policies promoting gender equality in the hiring and remuneration procedures for average employees do not boost R&D expenditure significantly. This partly explains the results in previous studies that found that female representation reduced involvement in risky projects [6,7]. Thus, gender equality policies should be coupled with a "critical mass" of female workers in R&D teams to significantly enhance a firm's innovation efforts.

Our study also contributes to the existing literature on employee compensation by demonstrating how gender equality policies for ordinary employees substantially influence a firm's investment decisions. Prior research has mostly examined the impact of executive compensation, specifically equity vega, on R&D expenditure [8]. However, our study highlights the significance of non-executive employee recruitment and compensation strategies as they also play a substantial role in influencing corporate investment. Innovation endeavors are primarily guided by specialist groups such as R&D teams and are subject to the influence of HR policies for non-executive personnel, in addition to executive compensation.

The remainder of this paper proceeds as follows. Section 2 presents a review of the related literature and our research hypotheses. Section 3 describes the data and empirical models used in this study. Finally, Sections 4 and 5 present our empirical results and conclusions, respectively.

2. Related Literature and Hypotheses Development

2.1. Determinants of Innovation Focus

Previous research on the determinants of a company's innovation focus has focused predominantly on financial capacities, industry dynamics, and technological infrastructure [3–5]. For example, prior studies have found a correlation between a firm's R&D expenditure and several factors such as the firm's ownership structure, cash holdings, liquidity, and volatility [3,9]. Additionally, a positive relationship exists between industry factors (e.g., market concentration) and average R&D expenditure [4]. A firm's R&D investment is also subject to the influence of technological infrastructure including state subsidies [5].

While the aforementioned factors are undeniably important, extant literature has focused predominantly on financial and industry-specific determinants and has afforded limited attention to the importance of "soft" assets accumulated within organizations in creating innovative activity. The difference in firm performance can be attributed to the long-term accumulation of specific internal resources and talents within the organization, according to the resource-based view [10,11].

Internal resources that promote a firm's innovation include not only tangible factors but intangible resources such as goodwill, patents, and HR [12]. If the value of a firm's HR is higher, the technical sensitivity and knowledge spillover effects may improve information absorption during the R&D process [13]. Additionally, Fleming (2001) implied that the employees of a firm with knowledge of the technological domains can boost the possibility of integrating knowledge to develop new technologies and R&D activities. Thus, such technical employees suggest that integrating the talents and knowledge within an organization may have a favorable impact on the enterprise's R&D activities [14,15].

2.2. Gender Equality HR Policies and Innovation Focus

With the active participation of women in the labor market in the modern era, gender diversity has emerged as one of the most important variables shaping the intellectual capital of businesses. To attract and retain talented women, many corporations are adopting gender equality HR policies such as paid maternal leave, equal employment opportunities for both genders, and pay equity between male and female workers [16]. Paid maternal leave refers to the paid time off work provided to new mothers after childbirth, with full or partial salary continuation. Affirmative actions include the provision of equal employment opportunities to ensure that all genders have the same access to jobs, promotions, and career development. Pay equity refers to the equal provision of compensation for similar work, regardless of gender.

While many organizations are recognizing the importance of gender equality in the workplace, there are still huge variations across corporations in terms of their pro-equality practices [17]. Moreover, in the literature, the specific impact of gender equality HR policies on a firm's innovation orientation remains an underexplored dimension within the finance and economics research. Existing literature, focused predominantly on financial- and industry-specific determinants [12], has afforded limited attention to the nuanced interplay between gender inclusivity initiatives and innovation pursuits. Consequently, there is a distinct gap in the understanding of how gender equality HR policies may influence a firm's innovation focus.

In recent years, the requirement to cultivate diversity and inclusivity within corporate environments has gained popularity as a critical driver of organizational performance. Several researchers investigating this topic have found that gender diversity in the boardroom positively influences firm performance [18,19]. Some other studies have reported a positive and nonlinear relationship between demographic diversity and performance [20]. Given that diversity can improve business performance, it places a stronger emphasis on long-term innovation and more sophisticated investment decisions in firms seeking to promote gender equality. For instance, diverse teams provide a wider range of perspectives, which can generate more comprehensive and innovative solutions [21]. Additionally, a gender-

inclusive work environment fosters a positive corporate culture, attracting and retaining top talent [22], ultimately enhancing the firm's overall performance and competitiveness in the market. This, in turn, facilitates the continuous pursuit of long-term innovation activities. Moreover, a diverse and inclusive environment encourages collaboration among team members with different perspectives [23], leading to the synthesis of unique and innovative solutions as well as the development of new approaches to problem-solving. Hence, gender equality policies create a conducive environment for diverse talent to thrive, which subsequently stimulates creativity, collaboration, and the generation of innovative ideas within a firm.

However, female representation in workplaces may sometimes deter involvement in innovative projects. Women are often perceived to be more risk-averse than their male counterparts [24]. Hence, women may be more inclined to thoroughly assess and mitigate potential downsides, leading to a more conservative approach to investment decision-making. Several studies have found that board diversity is associated with reduced investment risk [1]. This raises the empirical question of whether gender equality policies uniformly increase or decrease a firm's innovation focus. Using equal employment opportunities for both genders and pay equity as the empirical measures of gender equality HR policies, we examined their relationships with a firm's innovation focus as measured by R&D and intangible intensity. We formulated the following null hypothesis:

H1: *Gender equality HR policies are not associated with a firm's innovation focus.*

2.3. Does Workers' Gender Composition Matter?

The efficacy of gender equality HR policies is most prominent when a critical mass of employees benefits from their implementation. According to critical mass theory, there is a pivotal representation threshold that is particularly pertinent to marginalized groups such as female workers at which their influence becomes transformative, effecting substantial organizational change [25,26]. Consequently, initiatives that target the augmentation of female representation and the job satisfaction of female workers via gender equality policies will yield more significant organizational effects in contexts characterized by a greater proportion of female employees.

In the domain of corporate innovation, it stands to reason that female researchers in R&D roles can demonstrate their capabilities to a greater extent when bolstered by HR policies that espouse gender parity. For example, when female researchers are recruited for R&D positions without any gender-related bias in the hiring process, they are more likely to experience heightened levels of commitment and job satisfaction, which ultimately correlates with enhanced work performance. Furthermore, in instances where female R&D personnel receive equitable remuneration vis-à-vis their male counterparts, devoid of gender-related biases or disparities stemming from familial responsibilities, heightened levels of motivation can be anticipated, compelling a more dedicated pursuit of excellence in innovation-related pursuits. The capacity of female researchers to influence corporate innovation increases as their numerical strength attains a critical mass. Hence, a greater proportion of employees benefiting from gender parity provisions augments a firm's innovation-centric emphasis.

H2: *The relationship between gender equality HR policies and a firm's innovation focus becomes positive when there is a greater presence of female workers in R&D teams.*

3. Methods

3.1. Data

We derived the data used in this study from the Female Manager Panel Survey, conducted in the period 2020–2022. The Korea Women's Development Institute (KWDI), a government-supported research institute, conducts an annual survey of HR managers and female employees in Korean firms. The purpose of this survey is to gather information

on employment practices, remuneration techniques, and additional perks such as parental leave, which are particularly relevant to female workers. The survey also includes specific data regarding the employment and career advancement of female employees within organizations. Additionally, it includes the financial data of the employers of the survey participants. This provides a strong foundation for investigating the relationship between gender equality policies and innovation focus at the firm level.

The survey responses were collected using a combination of online surveys, mail surveys, and face-to-face interviews with managers and HR personnel working in companies with over 100 employees across the nation. The sample included female and male managers at the assistant manager level or higher as well as HR representatives, ensuring a diverse range of professional characteristics. Ethical considerations included informed consent and confidentiality measures, ensuring that all participants were aware of the study's purpose and their rights.

We utilized the firm-level responses provided by HR managers to ensure that their responses aligned with their firms' innovation activities. Our final sample, initially comprising 301 firm-year observations for the 2020–2022 period, was reduced to 130 observations (from 54 firms during the period 2020–2022). This sample excluded responses that did not include answers to necessary survey items and had missing financial variables. Table 1 displays the sample selection procedure and the sample distribution.

Table 1. Sample selection and distribution. Panel (**A**) is the sample selection; Panel (**B**) is the sample distribution by year.

Panel (A)	
Sample Selection	**Observations**
HR managers' responses to the survey during the 2020–2022 period	301
Less observations without responses to HR policy variables used in the analysis	(167)
Less observations with missing values of financial variables	(4)
Final sample	130
Panel (B)	
Year	**Observations**
2020	47
2021	41
2022	42
Total	130

This table reports the sample selection procedure (Panel (A)) and sample distribution by year (Panel (B)).

3.2. Empirical Model

To examine our first hypothesis, we estimated the following OLS regression of Equation (1) using 130 firm-year responses.

$$\begin{gathered} Innovation\ Focus_{i,t} = a_0 + a_1\ Equal\ Hiring_{i,t} + a_2\ Pay\ Gap_{i,t} + a_3\ Size_{i,t} + \\ a_4\ SalesGrowth_{i,t} + a_5\ Leverage_{i,t} + a_6\ ROA_{i,t} + a_7\ CFO_{i,t-1} + a_8\ Return_{i,t-1} \\ + Industry\ Fixed\ Effects + Year\ Fixed\ Effects + e_{i,t} \end{gathered} \quad (1)$$

where i and t denote the firms and years, respectively.

We utilized three proxies to measure a firm's innovation focus. The dependent variable, $Innovation\ Focus_{i,t}$, can be either $R\&D\ Intensity_{i,t}$ or $Intangible\ Intensity_{i,t}$. $R\&D\ Intensity_{i,t}$ is defined as the R&D capital (less depreciation) stated on the firm's statement of financial position, scaled by its total assets at the beginning of the year [27]. $Intangible\ Intensity_{i,t}$ is defined as the sum value of R&D capital, patents, industrial property rights, intellectual properties, and software assets (excluding their depreciations), scaled by the total assets at the beginning of the year.

In our research, the variables of interest were $Equal\ Hiring_{i,t}$ and $Pay\ Gap_{i,t}$. Both variables capture a firm's commitment to adopting gender equality HR practices. The value of $Equal\ Hiring_{i,t}$ is 1 if the firm has established specific targets or quotas for the recruitment and employment of women, and 0 if it has not. $Equal\ Hiring_{i,t}$ is an affirmative action measure intended to increase women's participation in the workforce and decrease gender-based discrimination in hiring practices [28]. $Pay\ Gap_{i,t}$ measures the difference in average compensation between men and women within a firm. In particular, it is defined as the difference between the natural logarithms of the average male compensation and the average female compensation. A higher $Equal\ Hiring_{i,t}$ value and a lower $Pay\ Gap_{i,t}$ value suggest that a company is more deeply committed to gender equality in its hiring and remuneration practices.

If the gender equality policies, on average, strengthen a firm's innovation focus, we expect the coefficient on $Equal\ Hiring_{i,t}$ to be significant and positive, and the coefficient on $Pay\ Gap_{i,t}$ to be significant and negative ($a_1 > 0$, $a_2 < 0$). In contrast, if gender equality policies, on average, weaken a firm's innovation focus, we expect the coefficient on $Equal\ Hiring_{i,t}$ to be significantly negative and the coefficient on $Pay\ Gap_{i,t}$ to be significantly positive ($a_1 < 0$, $a_2 > 0$).

While we included the industry and year fixed effects in our model to control for the industry- and time-specific invariant factors that drove our results, we further controlled for time-varying factors such as the firm size, growth options, financial structure, profitability, liquidity, and stock market information that may influence a firm's innovation activities [3,9,29]. $Size_{i,t}$ is the natural logarithm of the total assets of a firm at the beginning of the year. $SalesGrowth_{i,t}$ represents the growth rate of sales revenues in the past three years. $Leverage_{i,t}$ is defined as the total liabilities divided by the total assets, while $ROA_{i,t}$ is defined as the net income divided by the total assets. $CFO_{i,t-1}$ is the one year lagged cash flows from operation divided by one year lagged total assets. $Return_{i,t-1}$ is the buy-hold stock returns in the previous year.

To examine our second hypothesis, we estimated the following OLS regression of Equation (2), utilizing 130 firm-year responses.

$$\begin{aligned} Innovation\ Focus_{i,t} = {} & a_0 + a_1 Equal\ Hiring_{i,t} + a_2\ Equal\ Hiring_{i,t} \times R\&D\ Female\ Ratio_{i,t} \\ & + a_3 Pay\ Gap_{i,t} + a_4 Pay\ Gap_{i,t} \times R\&D\ Female\ Ratio_{i,t} + a_5\ R\&D\ Female\ Ratio_{i,t} \\ & + a_6\ Size_{i,t} + a_7\ SalesGrowth_{i,t} + a_8\ Leverage_{i,t} + a_9\ ROA_{i,t} + a_{10}\ CFO_{i,t-1} \\ & + a_8\ Return_{i,t-1} + Industry\ Fixed\ Effects + Year\ Fixed\ Effects + e_{i,t} \end{aligned} \tag{2}$$

where i and t denote the firms and years, respectively.

$R\&D\ Female\ Ratio_{i,t}$ is the proportion of female workers among the total number of R&D workers. To determine whether the relationships between gender equality policies and a firm's innovation focus varied with the proportion of female workers in R&D teams, we included interaction terms such as $Equal\ Hiring_{i,t} \times R\&D\ Female\ Ratio_{i,t}$ and $Pay\ Gap_{i,t} \times R\&D\ Female\ Ratio_{i,t}$. According to Hypothesis 2, we expected the relationship between $Equal\ Hiring_{i,t}$ and $Innovation\ Focus_{i,t}$ to become more positive as the value of $R\&D\ Female\ Ratio_{i,t}$ increased ($a_2 > 0$). Similarly, we expected the relationship between $Pay\ Gap_{i,t}$ and $Innovation\ Focus_{i,t}$ to become more negative as the value of $R\&D\ Female\ Ratio_{i,t}$ increased ($a_4 < 0$). The definitions of the other variables were the same as those described above.

3.3. Descriptive Characteristics

Panel A in Table 2 presents the descriptive statistics of the variables utilized in this study. In our sample, on average, R&D capital and intangible assets accounted for 0.8% and 1.2% of the total assets, respectively. Affirmative action was used by 8.5% of our sample to develop explicit hiring targets or quotas for the employment of women ($Equal\ Hiring_{i,t}$). The average pay gap was 0.114 and the average firm size was 13.984. Sales revenue growth was averaged at 11.3% during the last three years, while $Leverage_{i,t}$ had a mean value of 0.42. The average $ROA_{i,t}$ was 0.7%, while the average $CFO_{i,t}$ was 4.4%. The average stock returns for the sample observations was 5.6%. $R\&D\ Female\ Ratio_{i,t}$ is the percentage of female

workers in the entire R&D workforce. We utilized this variable in our secondary analysis to determine whether the association between gender equality policies and innovation focus was affected by the amount of female representation in the R&D field. Definitions of the variables are presented in Appendix A. In Table 2, Panel B presents the Pearson correlation coefficients among variables.

Table 2. Descriptive statistics and correlation matrix. Panel (**A**) shows the descriptive statistics; Panel (**B**) Pearson correlation coefficients among variables.

Panel (A)

Variables	N	Mean	Median	p25	p75	Min	Max	STD
R&D Intensity$_{i,t}$	130	0.008	0.000	0.000	0.001	0.000	0.132	0.024
Intangible Intensity$_{i,t}$	130	0.012	0.002	0.000	0.007	0.000	0.136	0.025
Equal Hiring$_{i,t}$	130	0.085	0.000	0.000	0.000	0.000	1.000	0.279
Pay Gap$_{i,t}$	130	0.114	0.091	0.052	0.180	−0.178	0.347	0.091
R&D Female Ratio$_{i,t}$	130	0.290	0.239	0.112	0.461	0.012	0.808	0.206
Size$_{i,t}$	130	13.894	13.310	11.958	15.812	10.550	18.691	2.320
SalesGrowth$_{i,t}$	130	0.113	0.046	−0.176	0.329	−0.823	3.123	0.560
Leverage$_{i,t}$	130	0.420	0.394	0.262	0.593	0.060	1.041	0.207
ROA$_{i,t}$	130	0.007	0.018	−0.038	0.061	−0.250	0.175	0.092
CFO$_{i,t-1}$	130	0.044	0.044	0.005	0.089	−0.250	0.316	0.091
Return$_{i,t-1}$	130	0.056	−0.029	−0.193	0.156	−0.810	2.423	0.433

Panel (B)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) *R&D Intensity*$_{i,t}$	1.000										
(2) *Intangible Intensity*$_{i,t}$	0.968	1.000									
	<0.0001										
(3) *Equal Hiring*$_{i,t}$	−0.105	−0.126	1.000								
	0.236	0.153									
(4) *R&D Female Ratio*$_{i,t}$	−0.062	−0.030	0.206	1.000							
	0.485	0.735	0.019								
(5) *Pay Gap*$_{i,t}$	−0.103	−0.038	−0.092	−0.135	1.000						
	0.244	0.669	0.299	0.126							
(6) *Size*$_{i,t}$	0.016	−0.030	0.080	0.126	0.134	1.000					
	0.858	0.737	0.365	0.154	0.130						
(7) *SalesGrowth*$_{i,t}$	0.013	0.014	−0.264	−0.033	0.088	−0.132	1.000				
	0.884	0.874	0.002	0.713	0.319	0.136					
(8) *Leverage*$_{i,t}$	0.345	0.365	−0.027	0.050	0.101	0.611	−0.136	1.000			
	<0.0001	<0.0001	0.760	0.570	0.254	<0.0001	0.123				
(9) *ROA*$_{i,t}$	−0.164	−0.188	−0.221	−0.017	0.078	−0.156	0.374	−0.435	1.000		
	0.062	0.032	0.012	0.852	0.381	0.075	<0.0001	<0.0001			
(10) *CFO*$_{i,t-1}$	−0.113	−0.094	−0.300	−0.183	0.127	0.094	0.075	−0.066	0.288	1.000	
	0.202	0.287	0.001	0.037	0.150	0.286	0.399	0.455	0.001		
(11) *Return*$_{i,t-1}$	0.038	0.061	−0.137	−0.161	0.281	−0.122	0.204	−0.114	0.190	0.105	1.000
	0.667	0.491	0.120	0.067	0.001	0.166	0.020	0.195	0.030	0.235	

This table presents the descriptive statistics and correlation coefficients among the variables. Panel (B) reports the Pearson correlation coefficients among the variables, derived from 130 observations, with the *p*-values listed below the coefficients.

4. Empirical Results

4.1. Gender Equality Policies and Innovation Focus

Table 3 reports the estimation results of Equation (1). The results for the dependent variables, *R&D Intensity*$_{i,t}$ and *Intangible Intensity*$_{i,t}$, are shown in Columns 1 and 2. Here, the variables of interest were *Equal Hiring*$_{i,t}$ and *Pay Gap*$_{i,t}$. Columns 1 and 2 indicate that the coefficients on *Equal Hiring*$_{i,t}$ were not statistically significant. The coefficient on *Pay Gap*$_{i,t}$ was significant at 10% in Column 1 but insignificant in Column 2. Thus, affirmative action to provide equal employment opportunities for both genders (*Equal Hiring*$_{i,t}$) and pay equity (*Pay Gap*$_{i,t}$), on average, do not present a clear relationship with a firm's innovation focus. Based on the results shown in Columns 1 and 2, we cannot reject the null hypothesis that gender equality HR policies are not associated with a firm's innovation focus. The adjusted R-squared values for columns 1 and 2 were 0.581 and 0.608, respectively, showing that the models had adequate explanatory power.

Table 3. Gender-equality HR policies and R&D intensity.

		Dep var:	
		(1)	(2)
VARIABLES	**Coef.**	***R&D Intensity***$_{i,t}$	***Intangible Intensity***$_{i,t}$
Equal Hiring$_{i,t}$	a_1	0.0068	0.0051
		(1.517)	(0.958)
Pay Gap$_{i,t}$	a_2	−0.0398 *	−0.0283
		(−1.925)	(−1.448)
Size$_{i,t}$	a_3	0.0025	0.0015
		(1.022)	(0.631)
SalesGrowth$_{i,t}$	a_4	−0.0023	−0.0027
		(−0.756)	(−0.935)
Leverage$_{i,t}$	a_5	0.0895 ***	0.0957 ***
		(2.676)	(2.900)
ROA$_{i,t}$	a_6	0.0527	0.0424
		(1.182)	(0.962)
CFO$_{i,t-1}$	a_7	−0.0001	−0.0002
		(−0.003)	(−0.014)
Returns$_{i,t-1}$	a_8	−0.0015	−0.0000
		(−0.253)	(−0.008)
Constant	a_0	−0.0702 ***	−0.0587 **
		(−2.715)	(−2.282)
Industry FE		Yes	Yes
Year FE		Yes	Yes
Observations		130	130
R-squared		0.688	0.709
Adjusted R-squared		0.581	0.608

This table reports the estimation results of Equation (1). Standard errors were corrected for heteroscedasticity. Corresponding t-stats are reported in parentheses. *, **, *** corresponds to the two tailed *p*-values < 0.10, 0.05, and 0.01, respectively. All variables are defined in Appendix A.

4.2. Worker Composition: Proportion of Female R&D Workers

Next, we identified and analyzed the possible factors that contributed to the overall lack of meaningful associations between gender equality legislation and a firm's emphasis on innovation. In particular, we examined the cross-sectional variations in the relationship between gender equality policies and the firm's innovation focus (i.e., whether the relationship becomes more positive when there is a greater proportion of female workers among the total number of R&D workers (*R&D Female Ratio*$_{i,t}$)). Table 4 reports the estimation results of the cross-sectional tests. The dependent variables in Columns 1 and 2 were *R&D Intensity*$_{i,t}$ and *Intangible Intensity*$_{i,t}$, respectively. In Column 1, the coefficient on *Equal Hiring*$_{i,t}$ × *R&D Female Ratio*$_{i,t}$ was significant and positive (a_2 = 0.0455, $p < 0.05$). This indicates that providing equal opportunities for both genders in the hiring process is associated with a firm's greater innovation focus, as measured by R&D intensity. In the same column, the coefficient on *Pay Gap*$_{i,t}$ × *R&D Female Ratio*$_{i,t}$ was significant and negative (a_4 = −0.1944, $p < 0.1$). This indicates that promoting gender pay equity in the remuneration process is associated with a firm's greater focus on innovation activities, as measured by R&D intensity.

In Column 2, wherein *Intangible Intensity*$_{i,t}$ is the dependent variable, the results demonstrated a similar pattern to those shown in Column 1. Thus, the findings presented in Table 4 are consistent with Hypothesis 2 and indicate a notable interaction effect. Specifically, these results suggest that the positive impact of gender equality policies on a firm's innovation focus is amplified when a larger proportion of employees benefit from these policies, particularly in the realm of R&D. This emphasizes the importance of both gender-inclusive HR policies and a balanced workforce composition in fostering a culture of innovation within organizations. The adjusted R-squared values for Columns 1 and 2 were 0.591 and 0.626, respectively, showing that the models had adequate explanatory power.

Table 4. Gender equality HR policies and R&D intensity in the increased representation of female workers in R&D.

		Dep var:	
		(1)	(2)
VARIABLES	**Coef.**	***R&D Intensity***$_{i,t}$	***Intangible Intensity***$_{i,t}$
Equal Hiring$_{i,t}$	a_1	−0.0105	−0.0181 *
		(−1.215)	(−1.850)
Equal Hiring$_{i,t}$ × *R&D Female Ratio*$_{i,t}$	a_2	0.0455 **	0.0615 ***
		(2.286)	(2.904)
Pay Gap$_{i,t}$	a_3	0.0268	0.0363
		(0.656)	(0.953)
Pay Gap$_{i,t}$ × *R&D Female Ratio*$_{i,t}$	a_4	−0.1944 *	−0.1913 *
		(−1.726)	(−1.845)
R&D Female Ratio$_{i,t}$	a_5	0.0367 **	0.0420 **
		(2.086)	(2.422)
Size$_{i,t}$	a_6	0.0024	0.0015
		(0.985)	(0.651)
SalesGrowth$_{i,t}$	a_7	−0.0034	−0.0044
		(−1.079)	(−1.556)
Leverage$_{i,t}$	a_8	0.0950 ***	0.1011 ***
		(2.803)	(3.031)
ROA$_{i,t}$	a_9	0.0521	0.0427
		(1.209)	(1.012)
CFO$_{i,t-1}$	a_{10}	0.0092	0.0099
		(0.592)	(0.663)
Returns$_{i,t-1}$	a_{11}	−0.0025	−0.0007
		(−0.406)	(−0.119)
Constant	a_0	−0.0811 ***	−0.0730 ***
		(−3.040)	(−2.758)
Industry FE		Yes	Yes
Year FE		Yes	Yes
Observations		130	130
R-squared		0.705	0.730
Adjusted R-squared		0.591	0.626

This table reports the estimation results of Equation (2). Standard errors were corrected for heteroscedasticity. Corresponding t-stats are reported in the parentheses. *, **, *** corresponds to the two tailed *p*-values < 0.10, 0.05, and 0.01, respectively. All the variables are defined in Appendix A.

5. Discussion

The findings of this study provide a nuanced understanding of the relationship between gender equality HR policies and a firm's innovation focus, measured through R&D and intangible asset intensity. Contrary to some prior research that suggested a straightforward positive impact of gender diversity on corporate innovation [1,2], our results indicate that gender equality HR policies alone do not universally enhance R&D intensity. Instead, the benefits of these policies become more pronounced when there is a significant presence of female employees in R&D roles.

This observation aligns with critical mass theory, which posits that the influence of marginalized groups such as women in the workforce becomes substantial when they reach a pivotal representation threshold [25,26]. Our study showed that gender equality policies are more effective in promoting innovation when a firm has already achieved a higher proportion of female employees within its R&D teams. This finding underscores the importance of not only implementing gender equality policies, but also ensuring that these policies are supported by a critical mass of female employees to realize their full potential.

6. Conclusions

This study examined the relationship between gender equality HR policies and a firm's innovation focus in a time that is characterized by the pressing need for both in-

novation and inclusivity. The findings highlight that gender equality policies including equal employment opportunities and gender pay parity are not significantly correlated with a company's dedication to R&D efforts, on average. More importantly, our research findings demonstrate that the relationship between gender equality policies and a company's inclination toward innovation is heightened when the R&D staff comprises a greater percentage of female contributors. The aforementioned amplification effect highlights the utmost significance of not only enacting gender-inclusive HR policies, but also in fostering a well-balanced staff composition, particularly in the R&D domain.

In light of our findings, this research carries significant practical implications. The amplified impact of gender equality policies on R&D teams with a higher female representation underscores the importance of having an equitable workforce composition in innovation-driven endeavors. In this study, we presented a multidimensional perspective on the interplay between gender equality policies, organizational innovation, and workforce diversity, contributing to the burgeoning discourse on gender-inclusive corporate strategies.

Our findings have broader ramifications outside the scope of gender equality campaigning. These ideas will resonate deeply with professionals in every field as well as everyone involved in decision-making processes, prompting the call for reassessing HR methods and tactics within organizations. Increasingly, the adoption of diversity and inclusivity, particularly within R&D divisions, is being recognized as a crucial strategic necessity for organizations seeking to enhance their competitive advantage in the current business environment.

Our study underscores the significance of gender-inclusive HR policies in fostering innovation within firms, particularly when there is a higher representation of female employees in R&D departments. This finding has several practical implications for organizations and policymakers. First of all, companies should prioritize the development and implementation of gender equality policies such as equal employment opportunities and pay parity, especially in R&D departments. These policies are not only ethical, but are also strategically beneficial for enhancing innovation. Secondly, firms can leverage a diverse workforce as a strategic asset. By fostering an inclusive environment where female R&D staff can thrive, companies may enhance their innovation capabilities, leading to a competitive advantage in the marketplace. Third, policymakers and regulatory bodies can use these insights to advocate for and design policies that promote gender equality in the workplace. Monitoring the effectiveness of such policies in promoting innovation can help in refining them for better outcomes. Finally, organizations should invest in creating a culture that supports diversity and inclusion. Training programs aimed at reducing unconscious biases and promoting inclusive behaviors can be beneficial in maximizing the positive effects of gender equality policies on innovation.

In summary, this research contributes to the existing body of knowledge regarding the intersection of gender equality policies and innovation endeavors, providing insights into the complex factors required for the achievement of organizational objectives. By examining the complicated parts of this relationship, we contribute to the growing conversation about gender-inclusive corporate strategies, working toward a future wherein diversity, equality, and innovation form the pillars of organizational success.

However, several limitations should be acknowledged. First, our study was based on data from Korean listed firms, which may limit the generalizability of the findings to other cultural or regional contexts. Additionally, the study relied on survey data, which may be subject to self-reporting biases. Future research could benefit from a longitudinal approach to examine the long-term effects of gender equality policies on innovation outcomes and could also explore the impact of these policies in different cultural settings to enhance the generalizability of the findings.

Further avenues for research include investigating the specific mechanisms through which gender equality HR policies influence innovation such as the role of organizational culture, leadership support, and employee engagement. Additionally, future studies could

examine the interaction between gender equality policies and other forms of diversity such as racial or age diversity to provide a more comprehensive understanding of how inclusive HR practices can drive innovation. By addressing these areas, future research can build on our findings to develop more effective strategies for leveraging diversity to foster corporate innovation.

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Appendix A Variable Definitions

Variables	Definitions
$R\&D\ Intensity_{i,t}$	R&D capital (less depreciation) on the firm's statement of financial position, scaled by its total assets at the beginning of the year.
$Intangible\ Intensity_{i,t}$	Sum value of R&D capital, patents, industrial property rights, intellectual properties, and software assets (excluding their depreciations), scaled by the total assets at the beginning of the year.
$Equal\ Hiring_{i,t}$	Indicator that is equal to 1 if a firm has established specific targets or quotas for the recruitment and employment of women and 0 if it has not.
$Pay\ Gap_{i,t}$	Natural logarithm of the average male compensation minus the natural logarithm of the average female compensation.
$R\&D\ Female\ Ratio_{i,t}$	Proportion of female workers among the total number of R&D workers.
$Size_{i,t}$	Natural logarithm of the total firm assets at the beginning of the year.
$SalesGrowth_{i,t}$	Growth rate of sales revenues during the past three years.
$Leverage_{i,t}$	Total liabilities divided by total assets.
$ROA_{i,t}$	Net income divided by total assets.
$CFO_{i,t-1}$	One-year lagged cash flows from operation divided by the one-year lagged total assets.
$Return_{i,t-1}$	Buy-hold stock returns in the previous year.

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